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Oil Data Digest | Europe

Digging into the China Data

China has played a significant role in balancing the global oil market over the last few months. In this note we go through the latest data on China's oil trade, refining and demand sectors to understand how Chinese refiners were able to curtail imports by ~5 mb/d over March-May.

China has been a key balancing market in helping the world absorb the supply shock resulting from the closure of the Strait of Hormuz. Chinese refiners have cut crude imports dramatically over March-May, with total May crude imports at ~7.8 mb/d, down 3.2 mb/d YoY and almost ~5 mb/d lower than pre-conflict levels in February. However, observable crude inventory data shows China's crude stocks rising over the course of March and April before eventually drawing in May, which feels inconsistent with the large import cuts.

However, it must be recalled that prior to this conflict China was over-importing on crude for the majority of 2025 and early 2026, aggressively building both commercial and strategic oil inventories. This was true in February 2026, where China's crude balance appeared to be in a ~2.7 mb/d surplus, as crude imports outpaced refinery demand.

Cuts to crude imports therefore came against a backdrop of a strong supply surplus in China and does not mean oil demand has fallen one-for-one with imports. Focusing specifically on April, for which we have the full government dataset, domestic production and net crude imports of 4.4 mb/d and 9.3 mb/d respectively still marginally outpaced refinery demand at 13.4 mb/d. Observable crude inventories showing small builds in April now makes sense. Neither of these datasets are perfect, but directionally they tell the same story.

The drop in crude imports has been partly offset by lower refinery runs and likely depressed end user demand, particularly in petchems. Refinery runs in April were 1.8 mb/d lower than February and down 0.8 mb/d YoY. State-owned refiners have cut runs aggressively to minimise losses as high crude prices, refined product export restrictions and limited ability to pass through cost increases to domestic product prices had combined to turn refining margins negative across China by April. Independent refiners have instead shouldered more of the burden of supplying the domestic market, after being mandated by the government not to cut runs.

Our apparent demand calculation estimates China's underlying end-user oil demand to be running down 1.2 mb/d YoY. Demand declines YoY have been driven by LPG and fuel oil, as petchem facilities have cut run rates and independent refiners have shifted away from fuel oil as a feedstock. Diesel and gasoline demand appears marginally more resilient, as strong mobility data is offset by accelerating NEV sales. However, weak domestic prices do point at an overall soft transport fuel market.

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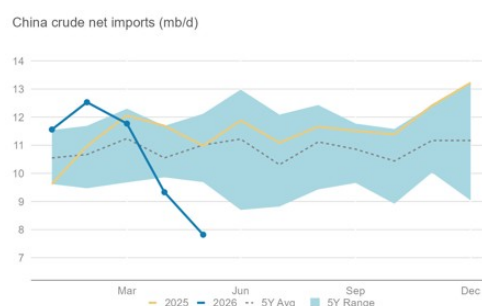
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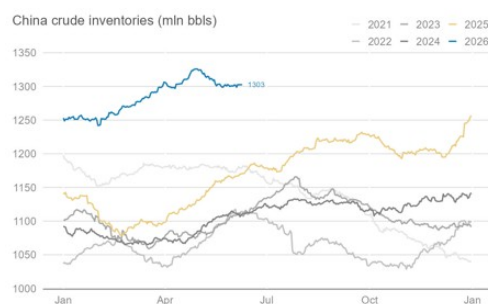
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Exhibit 1 : Crude net imports fell 2.4 mb/d MoM in April to 9.3 mb/d. Net imports then declined to 7.8 mb/d in May, down 3.2 mb/d YoY.



Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 2 : Observable Chinese crude inventories continued to build over March and April, based on Vortexa data. Average build rate was ~1.1 mb/d. However, inventories switched to draws in May, with an average draw rate of 750 kb/d.



Source: Vortexa, Morgan Stanley Research

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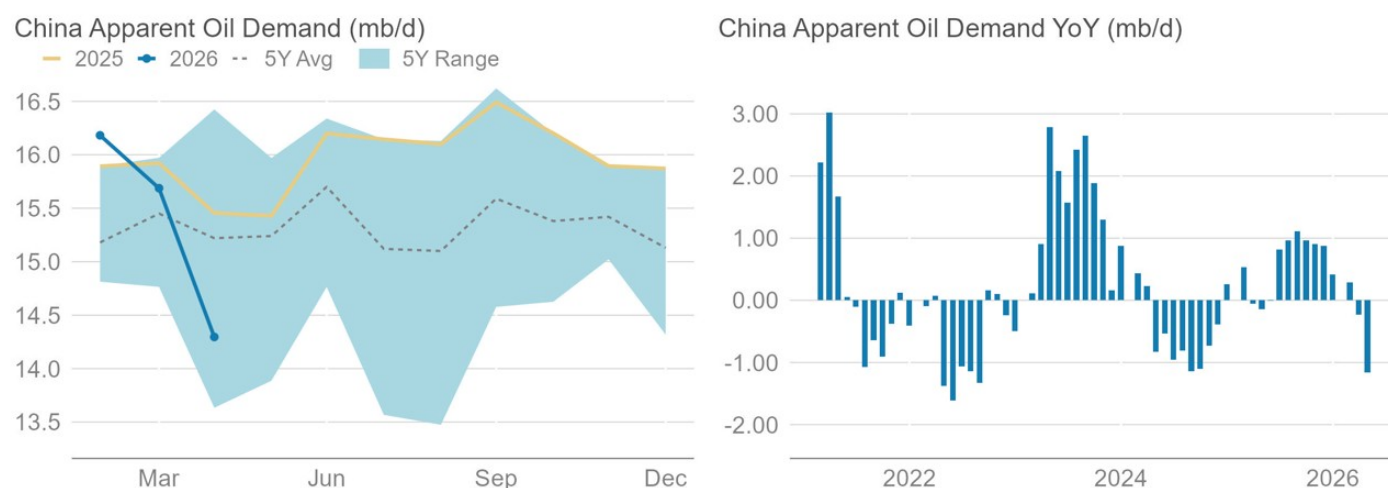
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China Oil Demand Data

China's total apparent oil demand was 1.2 mb/d lower YoY in April, down 8%. Our apparent demand assessment is based on a combination of domestic crude production, refinery runs and net refined product trade data. This gives us an estimate for total refined product supply into the market. Actual consumption is difficult to measure and the lack of comprehensive refined product inventory data means we cannot easily split out final consumption from inventory draws/builds. Consequently, we refer to this assessment as 'apparent demand' rather than a definitive measure of actual consumption.

The drop in apparent demand in April was largely driven by a decline in refinery runs in China, with net refined product exports broadly flat YoY, given both exports and imports were ~500 kb/d lower YoY. Taking the individual calculations for apparent demand by product, a sharp decline in fuel oil and LPG demand appears to explain ~70% of the decline in total apparent oil demand YoY.

Exhibit 3: China's apparent oil demand fell 1.4 mb/d MoM and was 1.2 mb/d lower YoY (-8%) in April



Source: China General Administration of Customs, Morgan Stanley Research

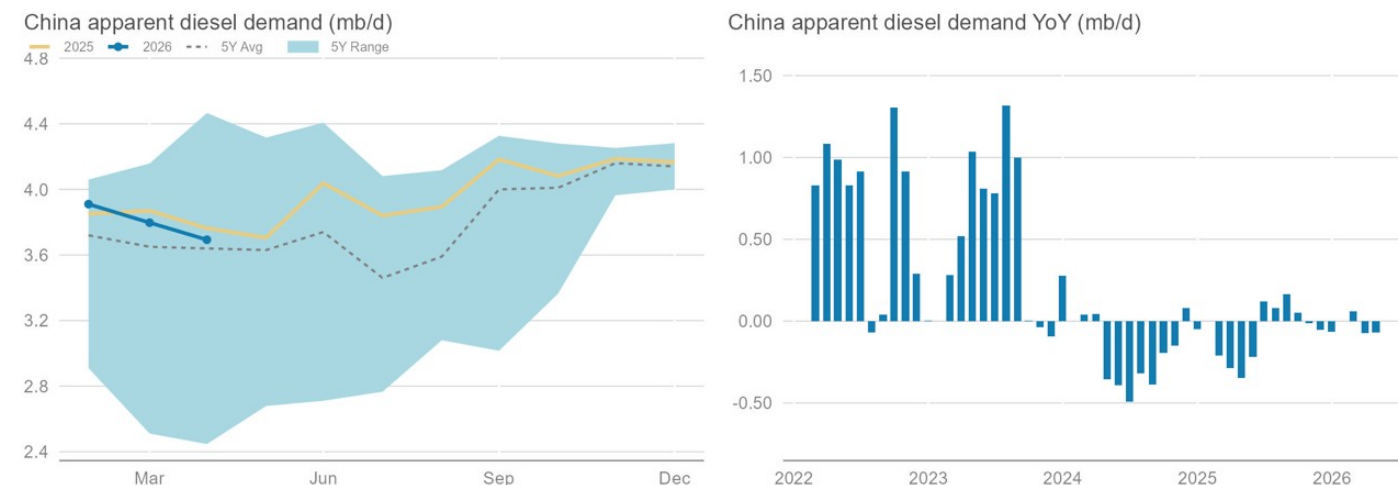
Apparent diesel demand in China fell to 3.7 mb/d in April, down 2% YoY. Apparent diesel demand in April averaged 210 kb/d lower than pre-conflict demand in February, although on a YoY basis diesel demand is only down 70 kb/d, indicating soft demand but not widespread demand destruction. Manufacturing activity declined over March to May, however still remained higher than a year ago. This firm industrial demand for diesel helped offset declines in the transport sector. In general, road fuel consumers in China have been partly protected from price increases. The government has limited pass through of higher refined product prices from refiners to consumers in China, only raising retail prices but half of what would usually have been passed through under the pricing mechanism. This may help explain the relatively small decline in apparent demand YoY.

Accelerated deployment of LNG and NEV heavy trucks is eroding diesel demand in China, driven by rising oil prices and expectations of firm trucking demand at the start of 2026. Although retail diesel price rises have been capped, high fuel price volatility and expectations of future price hikes is still incentivising switching to NEVs or LNG trucks at the margin. Argus estimates that diesel displacement by new energy and LNG heavy trucks increased by 230 kb/d on the year in January-March, based on mileage data. LNG trucks remain an attractive prospect, as LNG is still the cheaper fuel on a energy equivalent basis, with LNG heavy truck penetration in sales reaching 26% in April. China's domestic electric heavy truck sales grew by over 90% YoY in May, taking estimates of domestic penetration to ~40% in May, up from ~30% sales penetration in January (CVWorld, June 2nd). Expectations of a stronger manufacturing sector in 2026, as the uncertainty from US tariffs faded, boosted expectations of trucking demand and prompted heavy truck fleet build out into 2026.

China's manufacturing PMI peaked in March but then saw two months of consecutive decline over April and May, indicating weakening manufacturing activity. Slowing new orders was a driver of the declining PMI prints, partly reflecting business uncertainty

around impacts from events in the Middle East. A slowdown in fiscal spending has also weighed on new orders, with construction business activity a noticeable laggard. However, despite the weakening trend, PMIs remained in expansion territory (>50) over March and April before dropping back to 49.9 in May. PMIs were also higher than April and May 2025 levels, when the US-China trade war weighed on business activity, indicating improved industrial activity YoY. New orders for export specifically remained strong in Mar-Apr, likely reflecting strength in key energy-transition related exports, such as EVs, batteries and solar products amid the energy shock (see [here](#)).

Exhibit 4: Apparent diesel demand fell 100 kb/d MoM and was down 70 kb/d YoY (-2%) in April

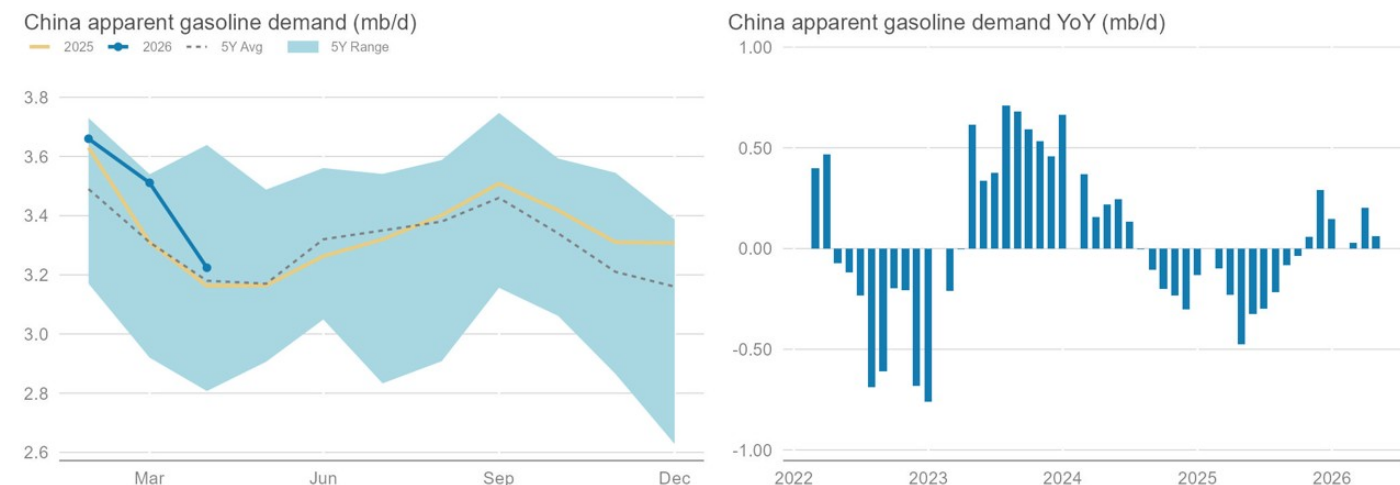


Source: China State Statistical Bureau/General Administration of Customs, Oilchem, Morgan Stanley Research

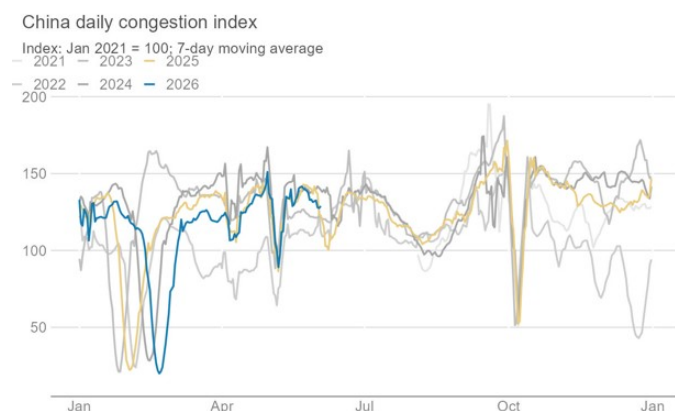
Apparent gasoline demand fell 290 kb/d MoM in April to 3.2 mb/d. Although gasoline demand appeared 2% higher YoY, April 2025 represented a very weak base. Meanwhile, the strong growth YoY in apparent gasoline demand observed over Jan-Feb has largely been eroded. NEV deployment continues at pace in the passenger sector in China, with data from the China Passenger Car Association showing sales penetration reaching 51% in March and 61% in April, up from 39% in January. ICE vehicle sales meanwhile have shrunk 37% YoY, dragging down overall numbers of passenger vehicle sales in April.

Argus estimates that Chinese gasoline demand is relatively insensitive to small price rises up to the +\$5-7/bbl threshold, at which point demand destruction accelerates. By the end of April, retail prices for gasoline had risen by \$30/bbl, passing through the sensitivity level of Chinese consumer. One independent refiner in Shandong with an integrated retail arm says its pump sales were around 20% lower in March-April compared with February (Argus 30th).

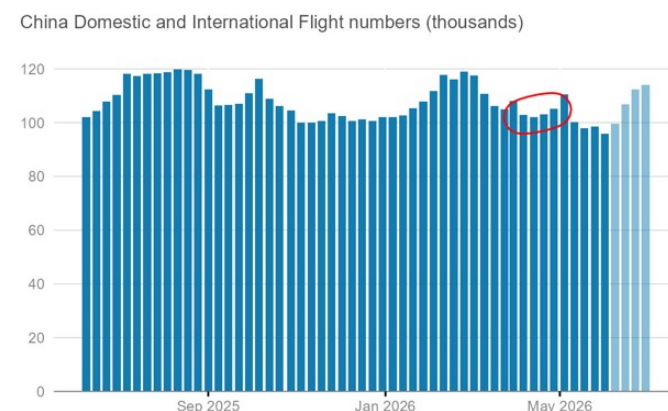
Into May, record EV use likely capped upside to gasoline demand despite record travel numbers for the Labor Day May holiday in China. Cross-regional passenger trips during the 1-5 May holiday rose by 3.5% YoY to a record 1.52 bln, Ministry of Transport data show. However, this appears to have not translated into higher retail gasoline sales, as rising EV and high-speed train use is weakening the link between public holidays and gasoline consumption. National Energy Administration data shows highway charging volumes for NEVs rose by 53% YoY during the 1-5 of May, with the transport ministry estimating that NEVs would account for around 24% of vehicles on highways during the holiday. Meanwhile, some gasoline retailers reported holiday sales were down by around 10-30% from a year earlier, with declines reaching as much as 50% in some of the more developed eastern and southern regions (Argus May 7th). This is despite independent refiners in Shandong cutting wholesale gasoline prices ahead of the holiday, in order to spur de-stocking.

Exhibit 5: Apparent gasoline demand fell 290 kb/d MoM but was up 60 kb/d YoY (+2%) in April

Source: China State Statistical Bureau/General Administration of Customs, Morgan Stanley Research

Exhibit 6: Traffic levels in China have tracked broadly flat YoY over May

Source: BNEF, Baidu, Morgan Stanley Research

Exhibit 7: China flight numbers were down 1% YoY in April, driven by a decline in domestic flight numbers.

Source: BNEF, Morgan Stanley Research

Apparent jet fuel demand fell 200 kb/d MoM in April, falling broadly flat YoY. Jet fuel demand is down 410 kb/d vs February levels vs a usual seasonal Feb-Apr decline of 210 kb/d. However, on a YoY basis jet fuel demand appears broadly flat, indicating flight growth has been curtailed but there has not been destruction of base jet fuel demand yet.

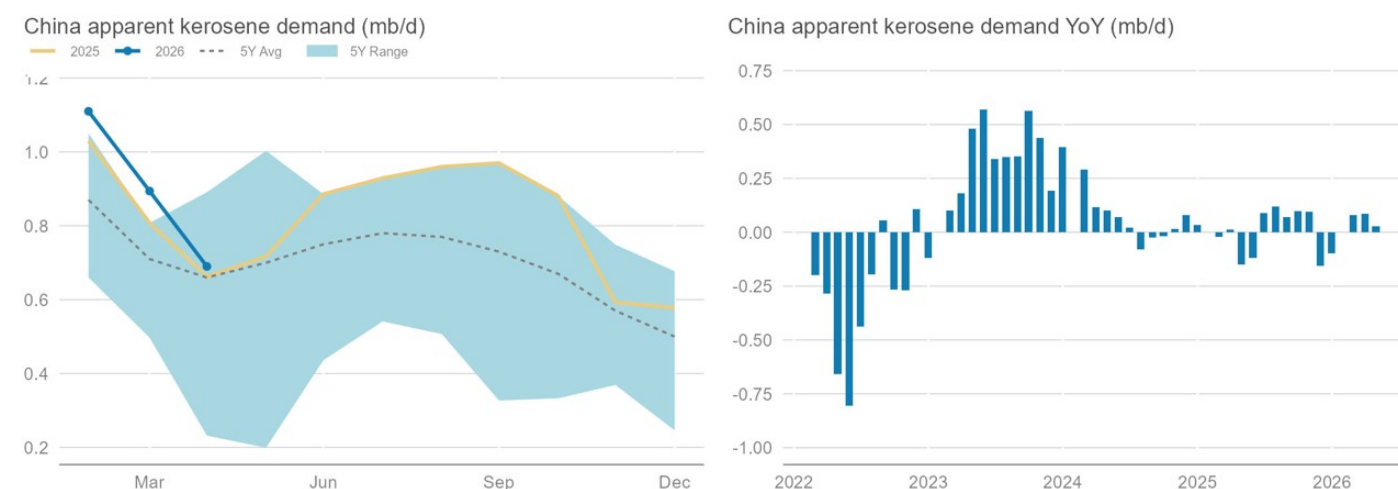
Rising jet fuel costs since April have pushed Chinese airlines to cancel flights, cutting jet fuel demand. Jet fuel ex-refinery prices rose by \$78/bbl MoM for April sales from refiners to China National Aviation Fuel (CNAF), the country's sole jet fuel trader, with the full increase passed through to airlines, pushing airlines towards breakeven margins. The NDRC also relegates how much of the cost increase airlines can pass through to ticket prices, permitting airlines to charge more for international tickets but capping ticket price increases for domestic flights. The airlines have responded by cutting short-haul domestic flights, where air travel already faces strong competition from high-speed trains. The national transport ministry reports that air passenger traffic during the May Golden Week holiday fell 6% YoY, while rail travel rose by 5% YoY. However, going forward flight cancellations should not worsen as the NDRC has intervened to prevent refiners from raising jet fuel ex-refinery prices for sales to CNAF again in May.

Cuts to domestic flight numbers are also visible in BNEF's flight tracker, which shows a 1% decline YoY in domestic flight numbers in China in April. Domestic flight numbers declined by 7% MoM in March followed by a 6% MoM decline in April. A drop in March flight numbers is considered normal, as travel slows down post the Lunar New Year holiday, however April usually records positive growth vs March flight levels. Looking forward, domestic flight numbers declined by 4% YoY in May as flight cancellations grew.

International flight numbers continued to grow, recording a 4% rise YoY in April. However, this is a slower rate of growth than usual, with international flight numbers growing at an average rate of 10% YoY in 2025. Looking forward, international flight growth compressed even further in May and June, with flight numbers broadly flat YoY. This will continue to weigh on jet fuel demand.

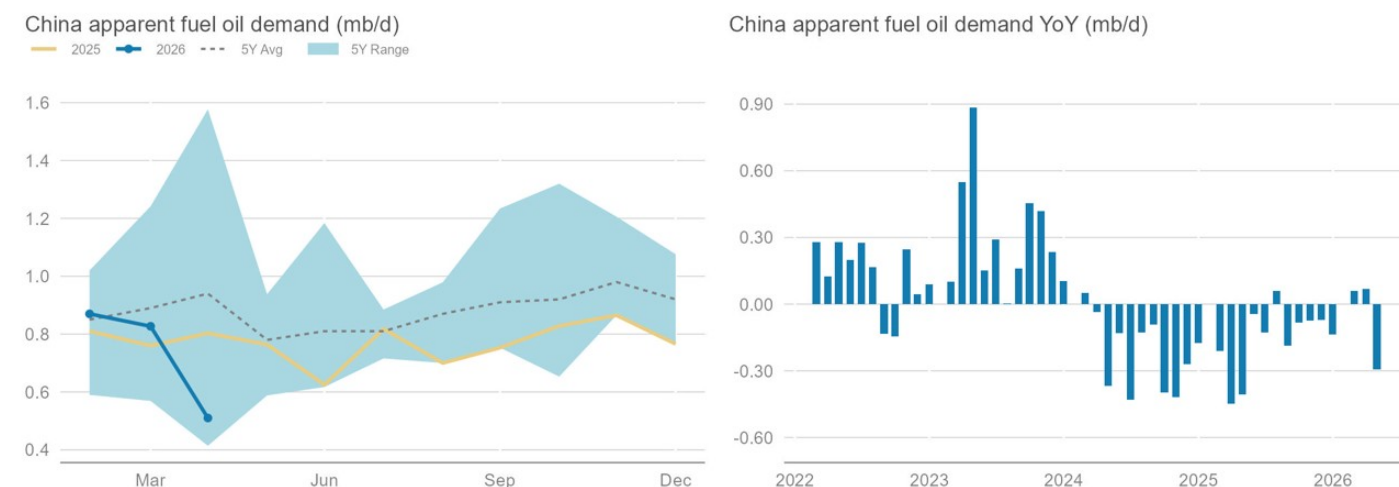
Our adjusted estimate for jet fuel demand shows a 70 kb/d YoY decline, indicating more demand weakness than the pure apparent demand calculation from government data. The government data for jet fuel exports can skew the impression of demand, due to jet fuel for bunkering demand for international flights being included in the export category. Consequently, refuelling of international flights is not captured in demand. We can use tanker tracking data from Vortexa to adjust the export figure, so that only seaborne exports are included. On this calculation, apparent demand declined by 70 kb/d YoY in April vs 30 kb/d of growth for the non-adjusted figure.

Exhibit 8: Jet fuel/kerosene apparent demand fell 200 kb/d MoM but was up 30 kb/d YoY (+4%) in April



Source: China State Statistical Bureau/General Administration of Customs, Morgan Stanley Research

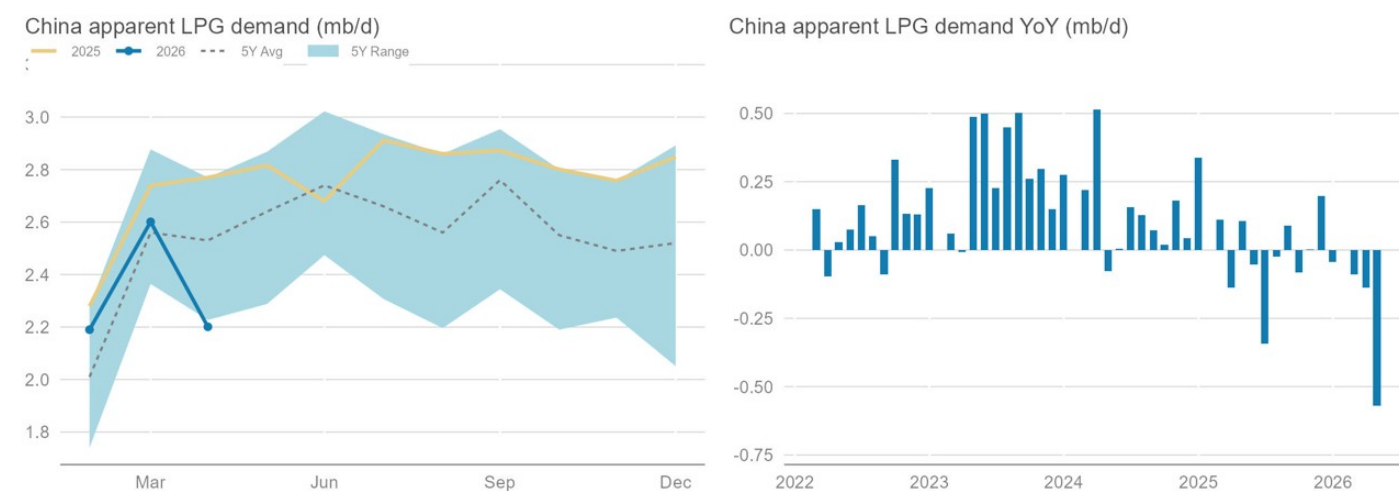
Apparent fuel oil demand dropped sharply in April, falling 315 kb/d to 510 kb/d. On a YoY basis, fuel oil demand was down 37%. The drop in apparent demand was driven by lower fuel oil imports. China's major fuel oil importers are the independent refiners, who often run fuel oil as a feedstock as an alternative to crude. Independent refiners are constrained on crude imports by the government run quota system, so processing fuel oil as well gives them more flexibility. China's three major fuel oil suppliers are Singapore, Russia and Iran. Iranian fuel oil exports have been limited by the US blockade, meanwhile volumes from Singapore were only 20% of pre-conflict levels in April. Russia remains a key fuel oil supplier; however, increased competition for Russian volumes following the temporary US sanctions-related authorisation for Russian oil-on-water reduced its attractiveness, driving down imports in May.

Exhibit 9: Apparent fuel oil demand fell by 315 kb/d MoM and was down 295 kb/d YoY (-37%) in April

Source: China State Statistical Bureau/General Administration of Customs, Morgan Stanley Research

Apparent LPG demand fell by 400 kb/d MoM in April, down almost ~300 kb/d YoY. LPG crackers and PDH units kept run rates relatively stable for the first several weeks of the Middle East conflict, supported by propane inventories. Run rates declined by only 3pp from late February to mid-March to 61%. This allowed plants to profit from the rise in downstream prices. Consequently, March demand for LPG was relatively firm, falling only 5% YoY.

However, disruptions to LPG imports became an issue after a few weeks, as plant inventories ran low. China relies on the Middle East Gulf for 75% of their seaborne LPG imports. Limited transit through the Strait of Hormuz since the start of the US-Iran conflict and delays at a US export terminal in March tightened global LPG supplies in April. China's LPG imports fell by 360 kb/d MoM to 600 kb/d as a result, down 50% YoY. Weaker margins also prompted refiners to bring forward maintenance and undertake run cuts, as downstream polypropylene prices have risen more slowly than propane feedstock costs. By the end of April, PDH run rates had dropped further to 47%, down 14pp from mid-March.

Exhibit 10: Apparent LPG demand fell by 400 kb/d MoM and was down 570 kb/d YoY (-21%) in April

Source: China State Statistical Bureau/General Administration of Customs, Morgan Stanley Research

Apparent naphtha demand fell by 75 kb/d MoM to 2.2 mb/d in April, down 60 kb/d YoY. Usually February marks the low point for naphtha demand in the year, as industrial facilities close during the Lunar New Year holiday. However, naphtha demand in China declined by 210 kb/d from February to April, indicating demand depression as a result of the Middle East conflict. March marked the first month of a YoY contraction in naphtha demand in China since April 2025, when US trade tariffs were introduced. Naphtha demand has otherwise

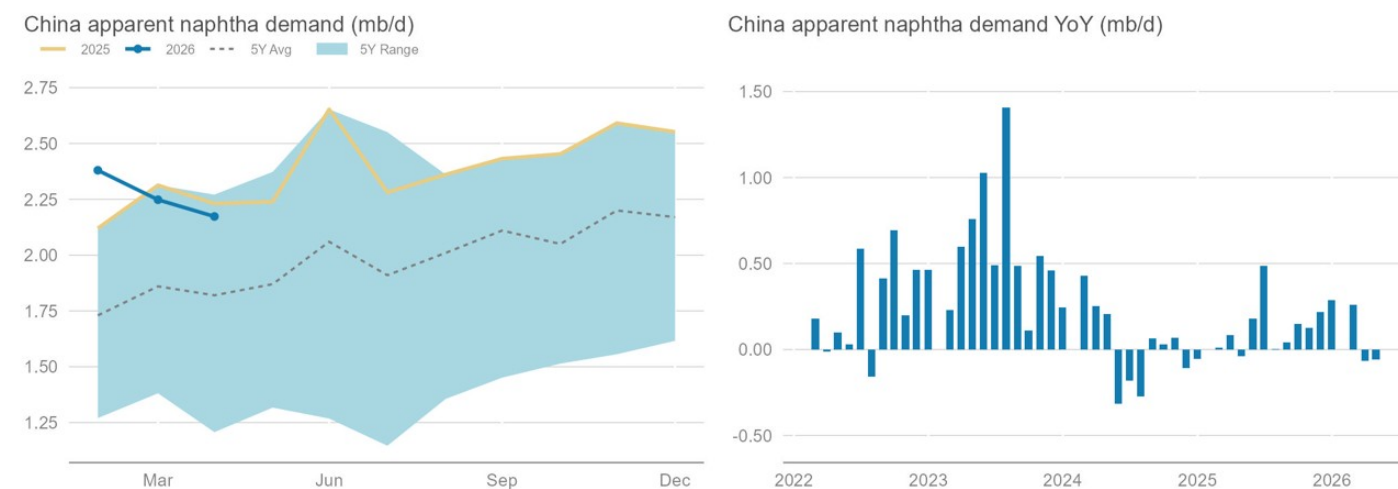
been growing strongly, supported by rapid expansion of China's steam cracker capacity and more diverse supply chain vs alternative feedstocks such as LPG and ethane, which are very US dependent.

However, despite greater naphtha supply chain diversity than LPG/ethane, China still relies on the Middle East Gulf for 45% of their naphtha supplies, the majority of which have been cut off since early March. China has managed to replace some of their naphtha supplies with alternative suppliers such as Russia and Algeria, however imports in April still averaged 40% lower YoY. High spot market prices for alternative supplies have also cut into petrochemical margins, with naphtha cracker cash margins averaging \$33/bl in March, prompting facilities to cut run rates. By early April, at least 18 steam crackers, representing ~22 Mtpa of capacity, had cut run rates. State-controlled Sinopec, PetroChina and Sinochem have all shut at least one steam cracker facility and cut run rates at several plants. Average steam cracker run rates in China dropped by 20pp from end-February to end-April, driving the 210 kb/d decline in demand from Feb-Apr.

Where possible, Chinese refiners have switched to processing ethane due to its more attractive margins. Around 13 Mtpa of China's 53 Mtpa steam cracking capacity can switch between ethane and naphtha feedstocks. The US is the sole major supplier of ethane to the global seaborne market, meaning this market has seen minimal supply disruption due to the closure of the Strait of Hormuz and delivered China prices were actually down 9% YoY in April. China's ethane imports hit an all-time high of 610 kb/d in April, double import levels in April 2025. Switching to ethane will likely continue going forward as long as the economics are more attractive.

Looking forward, some state-owned refiners raised petrochemical feedstock yields and ramped up steam cracker run rates in May on more attractive petrochemical feedstock margins vs transport fuels. Sustained steam cracker run cuts over March and April have led to PE and PX shortages in some regions, causing China's highly negative petrochemical margins to strengthen into May. Sinopec has now raised run rates at two of their steam crackers with PetroChina also following suit. However, the continued loss of seaborne LPG and naphtha supplies will likely keep the naphtha market tight and keep cracker run rates depressed YoY, as sourcing feedstock remains difficult. China is also scheduled to add 9.85 Mt of new cracker capacity to the market this year which was expected to be a major demand growth driver for naphtha. However, continued supply disruptions from the Middle East could result in start-up delays.

Exhibit 11: Apparent naphtha demand fell 75 kb/d MoM and was down 60 kb/d YoY (-3%) in April



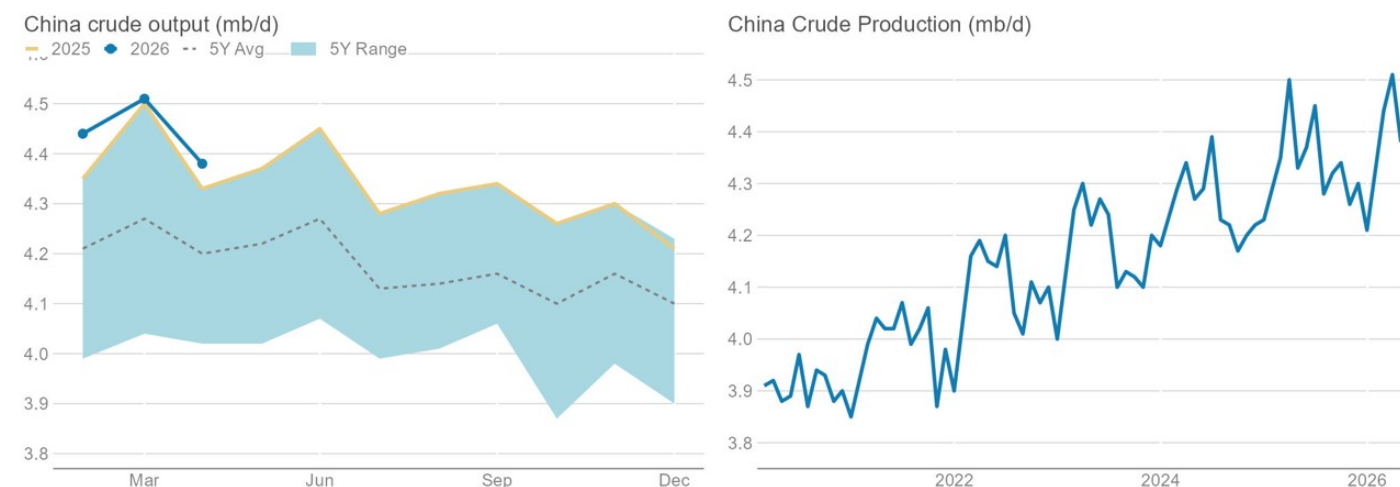
Source: China State Statistical Bureau/General Administration of Customs, Morgan Stanley Research

China Oil Production and Trade Data

Chinese crude production fell by 130 kb/d MoM in April to 4.38 mb/d, but still showed 1% YoY growth. Output decline MoM in April was led by Shaanxi and Gansu, partly offset by higher output from Guangdong and Shandong as new offshore fields ramped up,

China has been consistently growing its oil output on a YoY basis due to the start up of several new CNOOC fields since Q3 24. A recent milestone was the achievement of full production from Phase 1 of CNOOC's Kenli 10-2 oilfield in the southern Bohai Sea in late May. The project started producing in mid-2025 and has taken just under a year to ramp up to peak output of ~20.5 kb/d.

Exhibit 12: Crude production fell 130 kb/d MoM in April but was still up 50 kb/d YoY



Source: China State Statistical Bureau, Morgan Stanley Research

China's crude imports have collapsed over the course of March to May, as disruption to supplies from the Middle East alongside a rise in alternative crude grade prices has caused refiners to cut imports. China relies on cargos from the Middle East Gulf for 48% of their net crude imports and 35% of total crude demand, making the region a key supplier. March imports into China remained relatively resilient at ~11.8 mb/d, as the majority of this crude was procured over Jan-Feb and most Middle East cargos for March delivery had already set sail before the Strait closed.

Based on China NBS data, China's total crude imports (pipeline + seaborne) hit a 45-month low of 9.4 mb/d in April. The NBC data shows crude imports fell by 760 kb/d MoM in March to 11.8 mb/d, followed by a further 2.4 mb/d MoM decline in April to 9.4 mb/d. This was 2.3 mb/d (-20%) below April 2025 crude imports and shows a total decline of 3.2 mb/d vs pre-conflict levels.

Tanker tracking data from Vortexa shows a 3.5 mb/d decline in seaborne crude imports in April vs pre-conflict levels in February, a slightly larger decline than seen in the government data. Tanker tracking data has China's seaborne crude imports at 8.3 mb/d in April. The discrepancy in the level of import decline over Feb-Apr may be due to higher pipeline crude imports offsetting some of the loss in seaborne crude imports. This could show up in the NBS data but not the seaborne data. However, pipeline data from Argus only indicates a 30 kb/d uptick in pipeline crude import volumes from February to April, indicating there may be some other discrepancies such as differences between when China customs and Vortexa record a cargo as unloaded.

The Middle East accounted for about 77pc of the total monthly import decline in April, as crude imports were just 2.1 mb/d in April, roughly half of the levels in both March and April 2025. The decline in supply was largest from the UAE (-790 kb/d vs Feb), Iraq (-775 kb/d vs Feb) and Saudi Arabia (-310 kb/d vs Feb). Iraq relies heavily on the Strait of Hormuz for shipping, so was not able to redirect volumes via another route to China. The UAE faced less shipping constraints due to being able to use the ADCOP pipeline, however, tough competition for the remaining available Middle East barrels pushed up prices for these grades, discouraging Chinese buyers. Imports from Saudi Arabia were more resilient, as Saudi managed to divert 1.4 mb/d of crude supplies via Yanbu in the Red Sea to China to fulfill some of their term contract volumes.

Crude imports from Russia into China also declined by 420 kb/d from February to April, averaging 1.5 mb/d. The temporary US sanctions-related authorisation for Russian oil-on-water in the first week of March increased competition for Russian oil amongst Asian buyers and reduced discounts on Russian oil. The major buyers of Russian oil in China prior to the conflict were the independent refiners, who operate on thin margins and are very sensitive to the cost of crude. Argus also reports (March 6th) that Russian suppliers paused offers of Urals crude to China back in early March, on strong demand from Indian refiners who were offering to buy at a premium to Brent for ESPO and Urals grades in April. Discounts on Russia's ESPO Blend, a favourite of the independent refiners, moved from a \$8.5/bbl discount to Brent pre-conflict to \$6-8/bbl premium in April, causing Chinese refiners to cut back on purchases.

Crude volumes from Brazil and West Africa stepped up to partially offset China's supply gap. Imports of Brazilian crude averaged 1.3 mb/d in April, up ~390 kb/d vs April 2025 levels. However, imports of Brazilian crude were still ~170 kb/d lower than Feb 2026 levels, as China's state-owned refiners imported record volumes of Brazilian crude during the month due to its attractive prices and ability to be substituted for the Russian ESPO Blend that state-owned refiners had been cutting back purchases of since the Yulong refinery was sanctioned buy the US in October. Imports from Angola rose to 620 kb/d in April, nearly double Feb 2026 imports but marginally lower than imports in April 2025.

There are also reports that China was re-selling cargos of West African and Guyanese crude in April, following permission from the government for refiners to tap commercial crude stockpiles and weaker than expected domestic demand. These cargos were originally intended to be shipped to China, but were then offered by Chinese firms to Europe and other locations instead.

Imports of Iranian oil remained resilient over March and April, with Iranian oil cargos to China one of the few consistent types of transits seen through the Strait of Hormuz in the first two months of the conflict. Iranian volumes into China actually rose by 190 kb/d MoM in March to 1.8 mb/d, before dropping back to 1.4 mb/d in April. The temporary US sanctions-related authorisation for Iranian oil-on-water also helped drive stronger imports in March as state-owned refiners re-entered the market for Iranian oil and tapped oil-on-water volumes. This temporary authorisation expired on April 19, 2026 and was not renewed. Offers for Iranian Light flipped from discounts into premiums after the the temporary US sanctions-related authorisation, with Iranian Light was being offered at a \$2-3/bbl premium to Brent in April vs a \$10/bbl discount back in February. This weighed on imports by independent refiners in April and May, as the higher costs for Iranian crude squeezed their refining margins.

An increase in available VLCC berthing capacity also helped increase imports of Iranian oil in March. Several terminals in Rizhao, Shandong had suspended operations after being sanctioned by OFAC in October 2025, with no tanker unloadings recorded in Vortexa data for Nov-Dec 2025. This caused unloading delays with some tankers having to wait up to a month to discharge, driving a build up in Iranian oil-on-water off China of 30-40 mln bbls. However, in March several of these terminals began operating again, Argus reports (March 6th), allowing faster unloading of Iranian oil and unwinding the build up in oil-on-water.

Tanker tracking data indicates China's crude imports have slipped further in May, falling another 1.7 mb/d MoM to 6.6 mb/d. NBS data shows total imports in May fell to 7.8 mb/d. This is the lowest level of crude imports in over a decade and 3.6 mb/d lower YoY. Overall, China's crude imports have declined by 5.2 mb/d since the start of the Middle East conflict. A further reduction in imports of Russian and Iranian crude drove the MoM decline in May, as the US blockade hampered Iranian crude exports. Iranian arrivals into China still totalled 1 mb/d in May, although all of the cargos shown in Vortexa tanker tracking data had loaded in Iran in March or earlier i.e. before the US blockade began in mid-April.

Exhibit 13: Both Russia's Urals crude and Iran's Iranian Light crude grades priced at a premium to Brent over March and April, having priced at a discount since 2022 due to sanctions on these crudes.

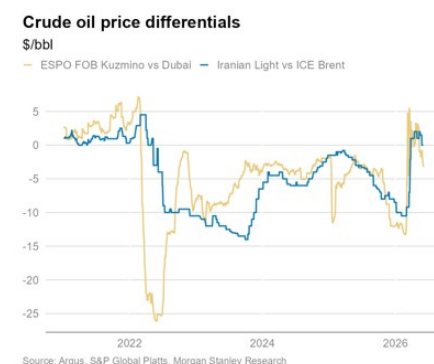
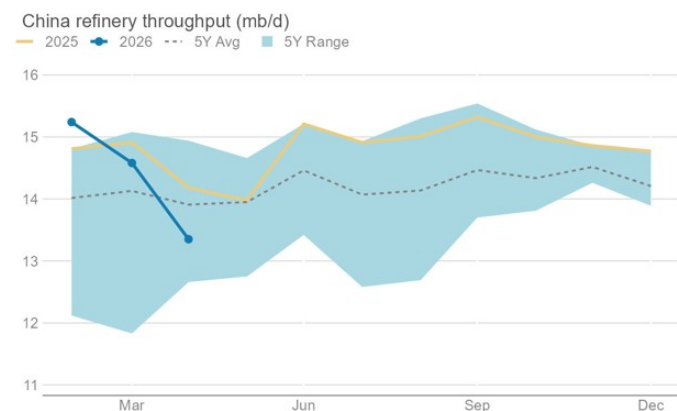
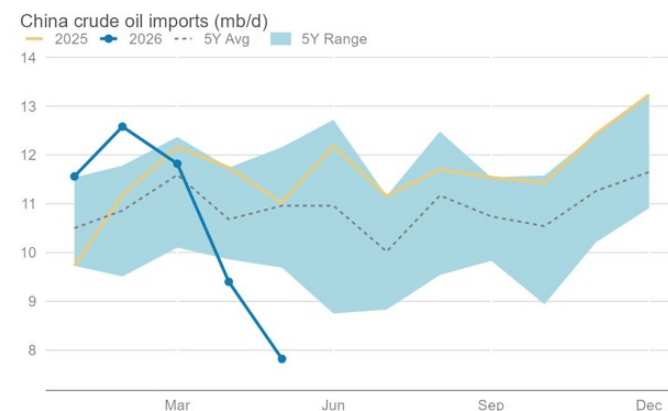


Exhibit 14: Refinery runs fell 1.2 mb/d MoM and were down 830 kb/d YoY (-6%) in April



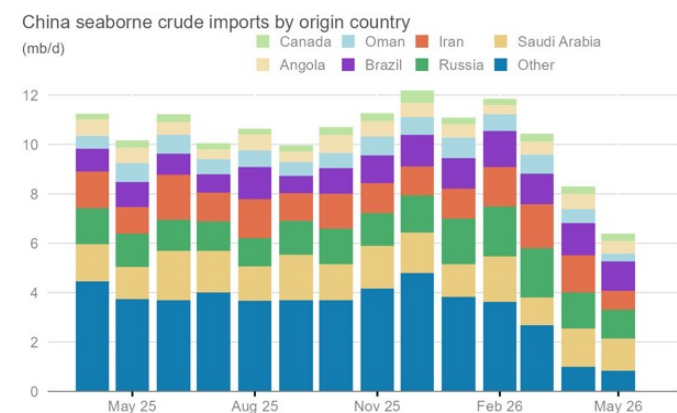
Source: China State Statistical Bureau, Morgan Stanley Research

Exhibit 15: Crude imports fell 2.3 mb/d lower YoY in April and were 3.2 mb/d lower YoY in May at 7.8 mb/d



Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 16: Seaborne data shows China's imports dropping to 6.6 mb/d in May, ~5 mb/d below February levels...



Source: Vortexa, Morgan Stanley Research

Exhibit 17: ...with Saudi share of imports remaining relatively stable as cargos were able to be re-routed via Yanbu



Source: Vortexa, Morgan Stanley Research

Exports

Introduction of restrictions on refined products exports from March has weighed on exports from China over April and May. On March 4th, China's NDRC issued a verbal request to exporters to stop signing new contracts and try to cancel shipments already sold but not yet loaded, in an attempt to prioritise supply to the domestic market amid disruptions to Middle East exports. Difficulties in sourcing crude, due to Chinese refiners' significant reliance on volumes from the Middle East, created concerns over future refinery run cuts and downside to transport fuel production. On March 11th the NDRC then issued a firmer instruction to refiners to halt all exports of transportation fuels (diesel, gasoline and jet fuel) that had not yet cleared customs, with immediate effect. Only bonded flows, mainly jet fuel used for international flights which is still reported as exports, were exempted alongside necessary flows to non-mainland areas (Hong Kong and Macau) and select neighbouring countries.

In response to the announcement of the restrictions on Chinese exports, regional product prices spiked. China plays an important role as a regional supplier, so restricting exports tightens the regional balance. This spiked export margins for Chinese refiners, however the export restrictions largely prevented Chinese refiners from capitalising on these elevated export margins. By late March, diesel export margins had reached \$75/bbl, while gasoline export margins were at \$20/bbl.

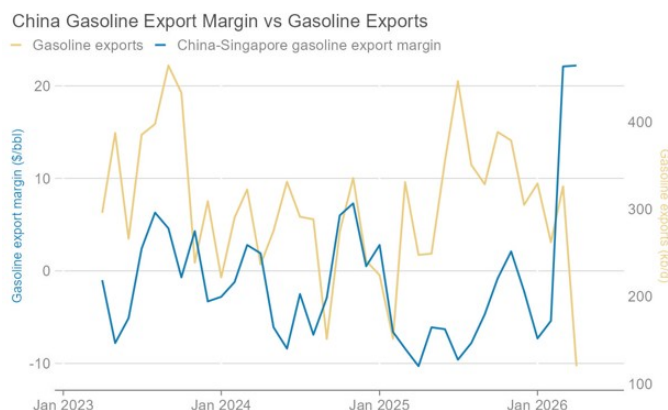
Despite the restrictions on refined product exports, Chinese refiners have been able to continue some exports of diesel and jet fuel. Although refined product export allowances under the export restrictions were only 50 kb/d for April, combined diesel, gasoline and jet

fuel exports appeared to average 250 kb/d, based on Vortexa tanker tracking data. The main recipients of these volumes were Singapore, the Philippines and Vietnam. Government data also shows higher levels of exports than the ban would necessitate, recording 70 kb/d of gasoline and diesel exports plus some jet fuel volumes. Nevertheless, volumes are down significantly vs pre-conflict, with April government data indicating diesel exports of 60 kb/d vs previous 12M average of 140 kb/d. Gasoline exports are down even further, with just 10 kb/d of exports in April vs previous 12M average of 150 kb/d. Jet fuel exports of 260 kb/d in April, including bonded volumes for international flights, are roughly 50% of pre-conflict flows. **Overall, refined product exports are down ~470 kb/d YoY in April, a 40% decline.**

In May, the government eased refined product export restrictions, granting additional special quotas for exports to designated "friendly" countries including Vietnam, Sri Lanka, Bangladesh, Myanmar and the Maldives. State-owned refiners have been permitted to export a combined 500 kt (around 125 kb/d) of oil products to these countries in May, with companies expecting to prioritise jet fuel exports. Total permitted exports for May are ~340 kb/d including bonded jet fuel exports and exports to non-mainland areas. This is still only half of usual export levels in 2025. Seaborne tanker tracking data from Vortexa indicates exports rose to ~400 kb/d in May, with a step up in exports to Australia, Bangladesh and Malaysia. Export margins were very positive in late April/early May (~\$70/bbl for diesel and ~\$20/bbl for gasoline), meaning refiners will have maximised exports where possible.

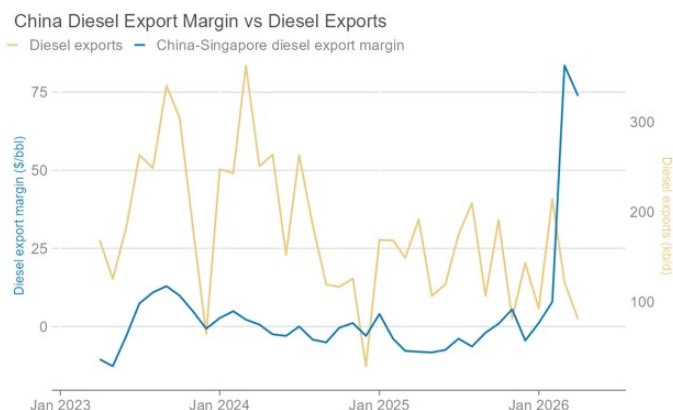
Looking forward, refined product exports are likely to continue tracking ~50% of usual levels in June, as the government keeps export controls in place while disruption in the Middle East continues.

Exhibit 18: Gasoline export margins averaged +\$22/bbl in March, when April loadings were scheduled



Source: Argus, Vortexa, Morgan Stanley Research

Exhibit 19: Diesel export margins averaged +\$84/bbl in March, when April loadings were scheduled



Source: Argus, Vortexa, Morgan Stanley Research

Imports

Total imports of refined products dropped by 540 kb/d MoM to 570 kb/d in April. This was half of the volume of imports recorded in April last year. Imports dropped across naphtha, fuel oil and LPG as disruptions to supplies from the Middle East and increases in product prices from alternative suppliers caused refiners to cut back on imports and reduce processing rates at both refineries and petrochemical facilities. China imports very small volumes (<50 kb/d) of transport fuels, relying on domestic refineries to produce diesel, gasoline and jet fuel. Instead, refined product imports are predominantly petrochemical feedstocks (LPG and naphtha) or fuel oil used as a refinery feedstock by independent refiners.

Imports of naphtha and LPG dropped sharply in April, down 60% and 37% MoM respectively. China usually receives 45% of their naphtha and 75% of their LPG seaborne imports from the Middle East Gulf, supplies that were largely cut off in March. On LPG, that flow usually equates to 40-50 LPG cargos a month, which fell to ~10 in each of April and May. The loss of imports was felt more significantly in April vs March, as the three week sailing time from the Middle East Gulf to China meant cargos were still arriving as usual until late March. Iran has remained the major supplier of LPG to China, continuing to export from the Gulf, albeit only delivering one-third of their usual monthly volumes pre-conflict. The US and Malaysia have been the other major suppliers of LPG to China over April and May.

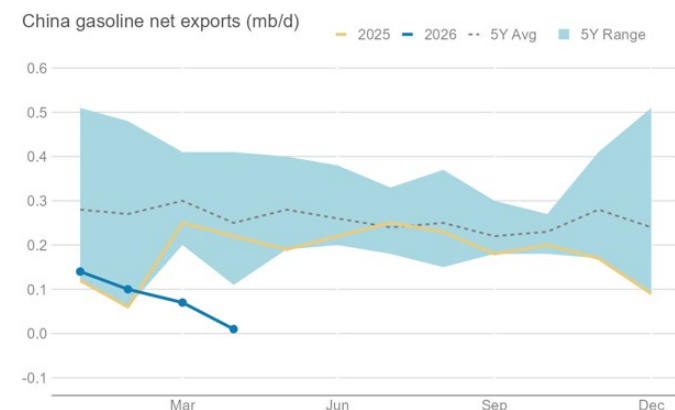
In the absence of naphtha supplies from the Middle East, Russia has emerged as the major supplier to China over April and May.

State-owned Sinopec resumed purchases of Russian naphtha after the US extended its sanctions license in mid-April, Argus reports (April 30th). However, Russian naphtha volumes were not significantly higher than pre-conflict as issues at Russian refineries have capped output of refined products. Consequently imports of naphtha by Chinese refiners in April of 190 kb/d were only ~40% of pre-war levels, as lost supply and higher naphtha prices discouraged purchases. Refiners have also been pivoting to ethane as an alternative feedstock where possible, as prices are more attractive given this market has relatively low exposure to the Middle East. However, given the majority of ethane exports come from the US, concerns around over-dependency on US supply will likely cap upside to ethane imports into China.

Imports of fuel oil have weakened significantly as negative refining margins, extra crude import quota and high fuel oil costs have weakened the economics of running fuel oil as a refinery feedstock. This is down from record February levels, which were boosted by Chinese refiners looked to replace lost Venezuelan crude imports with fuel oil for feedstock. Chinese refiners often declared Venezuelan heavy crude imports as diluted bitumen, meaning they fell outside the crude import quotas. Consequently swapping to fuel oil was an attractive option, as this also falls outside the import quota system. Volumes from Singapore, China's major source of supply, were only 20% of pre-war levels in April, as high prices discouraged buying. Independent refiners in China also usually purchase ~130 kb/d of Russian fuel oil which is often priced at a discount to other suppliers. However, similar to crude, the pool of willing buyers of Russian crude and refined products expanded in March as Middle East supplies fell, driving diversion of Russian fuel oil away from Chinese independent refiners to other Asian nations in May.

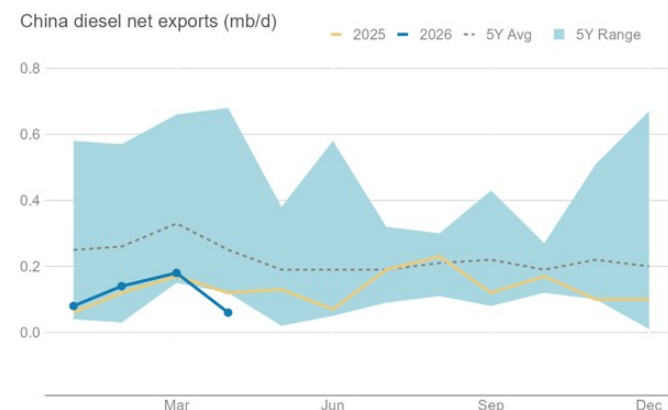
Looking forward, imports of LPG and naphtha remained depressed in May, as high regional prices disincentivised imports. Instead Chinese state-owned refiners raised output of LPG and naphtha from their own refineries where they could, in order to increase supply to petrochemical facilities and capture stronger margins, without raising imports. Refiners are also more likely to favour LPG over naphtha going forward, as propane has become more attractive on a delivered-Asia basis vs naphtha in May.

Exhibit 20: Gasoline net exports were down 60 kb/d MoM in April and 210 kb/d lower YoY



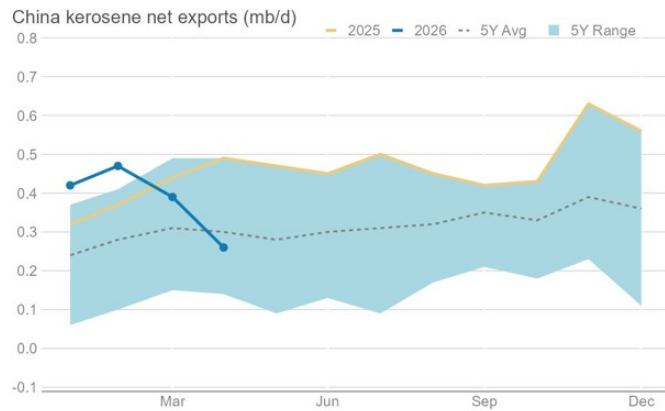
Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 21: Diesel net exports were down 120 kb/d MoM in April and 60 kb/d lower YoY



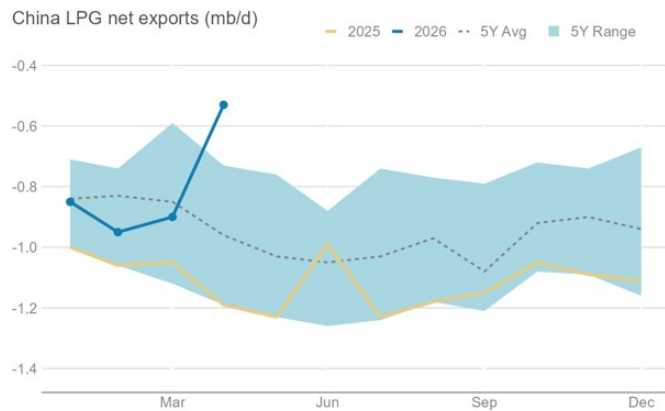
Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 22: Jet fuel net exports were down 130 kb/d MoM and 230 kb/d lower YoY



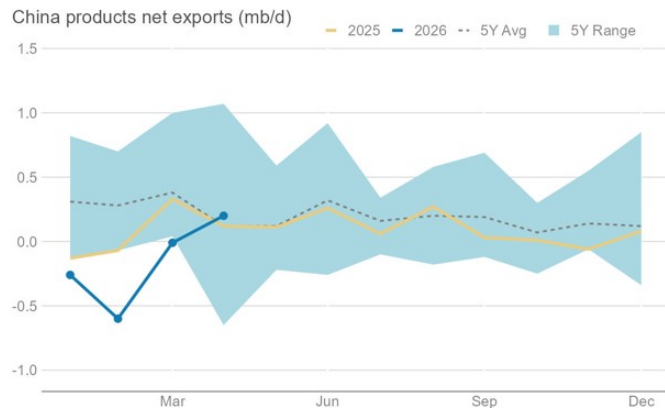
Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 24: LPG net exports were 370 kb/d higher MoM and 660 kb/d higher YoY



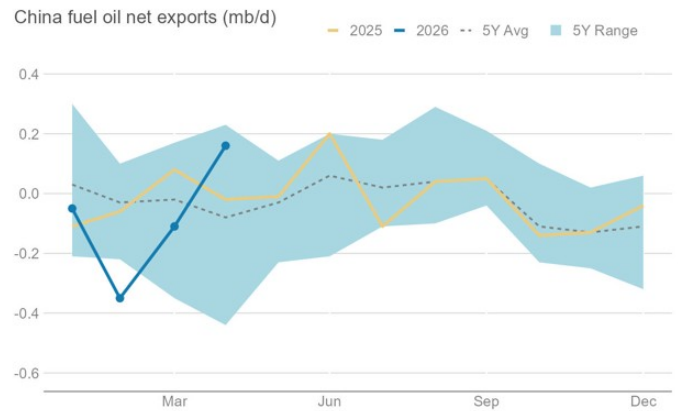
Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 26: Net refined product exports rose 210 kb/d MoM in April and were up 80 kb/d YoY



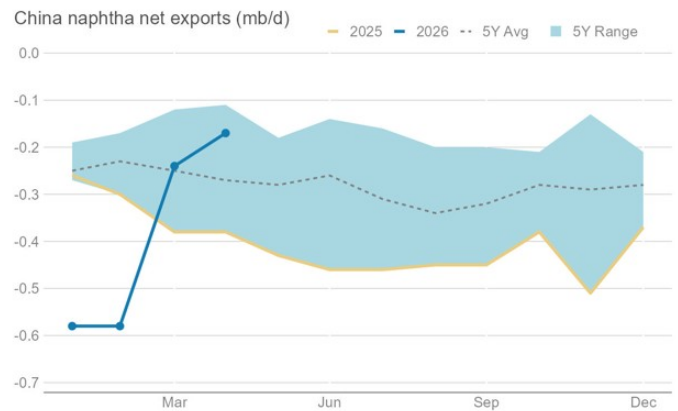
Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 23: Fuel oil net exports were up 270 kb/d MoM but 180 kb/d lower YoY



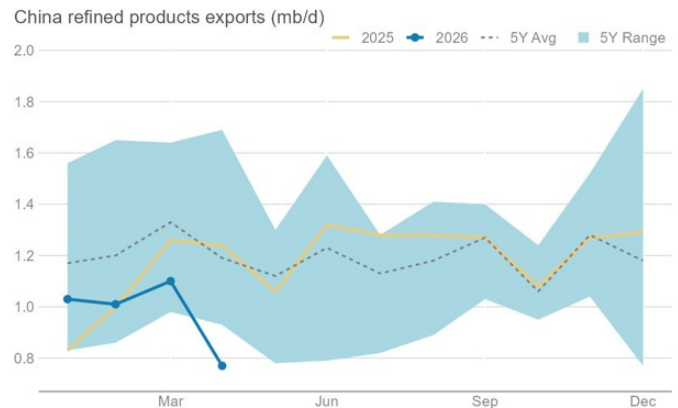
Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 25: Naphtha net exports were 70 kb/d higher MoM and 230 kb/d higher YoY



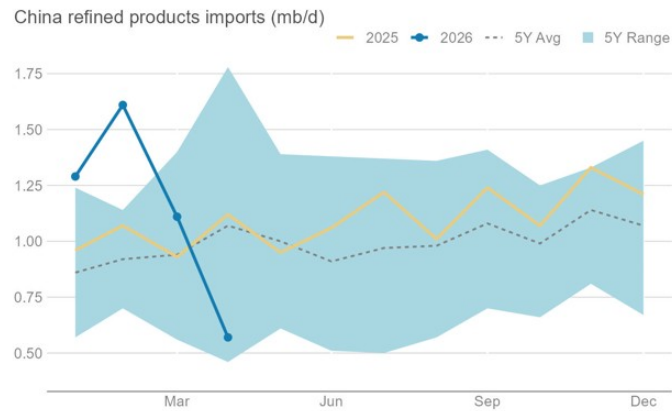
Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 27: Total product exports fell 330 kb/d MoM in April and were down 470 kb/d YoY (-40%)



Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 28: Total product imports were down 540 kb/d MoM in April and 550 kb/d lower YoY (-50%)



Source: China General Administration of Customs, Morgan Stanley Research

China Refining Data

Both gasoline and diesel cracks weakened significantly over March and April in China's domestic market, squeezing refining margins.

A sharp rise in crude costs for refiners combined with a more limited uptick in domestic refined product prices drove gasoline and diesel spot prices below the cost of crude in the domestic market. Diesel cracks fell to -\$11.8/bbl in April, down from \$12.1/bbl in February. Gasoline cracks dropped to -\$22.8/bbl in April from \$10.5/bbl in February.

Pump prices rose in the initial aftermath of the Middle East conflict, however these rises have lagged the rises in crude. This is because the National Development and Reform Commission (NDRC), which regulates pump prices for diesel and gasoline in China, declined refiners requests to pass through the full effect of the crude price spike. The NDRC raised retail prices for gasoline and diesel by \$27/bbl and \$30/bbl in its 23rd March and 7th April pricing cycles, only half the increase implied by the pricing formula that usually dictates sales prices.

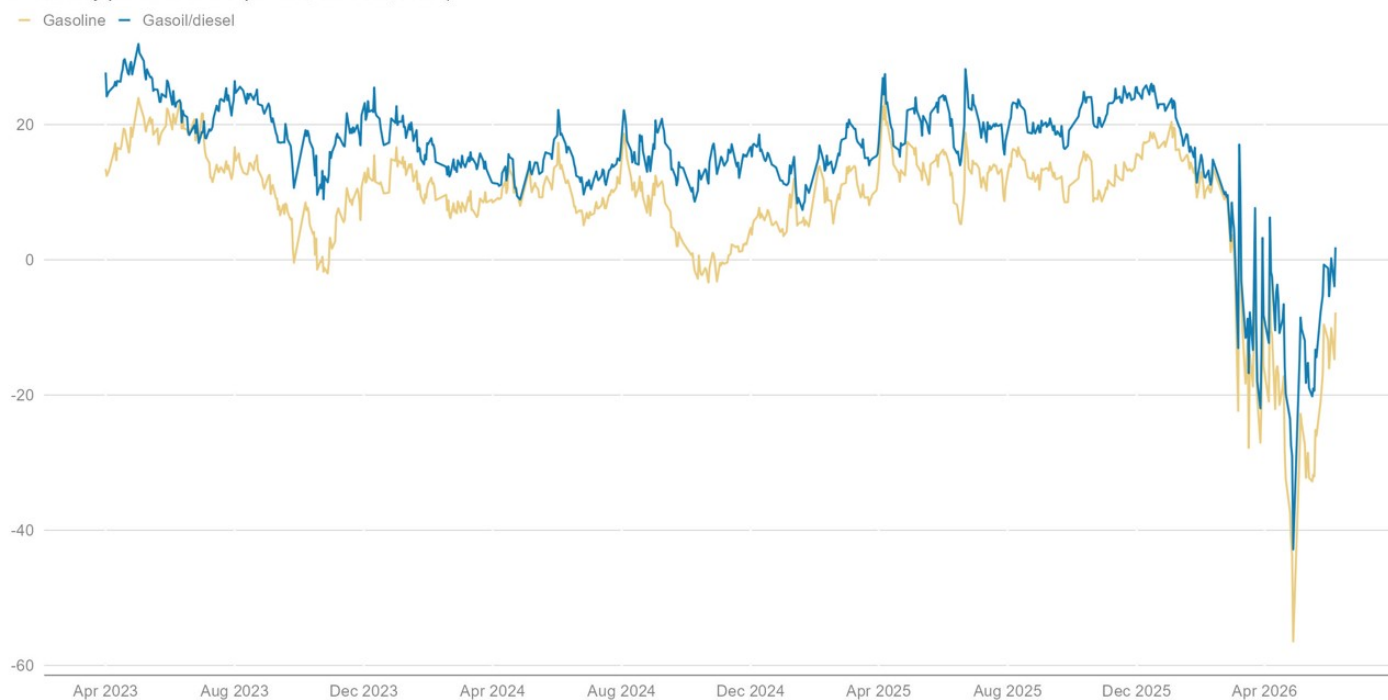
Despite these price protection measures, consumer demand for road fuels appears to have still weakened, driving lower gasoline and diesel sales and further weighing on domestic road fuel prices. Meanwhile restrictions on clean product exports from late March meant that state-owned refiners have had limited ability to capture high product prices in the tight overseas markets.

Looking forward, diesel and gasoline cracks both began to recover in 2H of May, but still remained negative. Easing crude prices and further rises in domestic product prices have marginally improved cracks in May, but overall cracks still averaged lower MoM during May. As of early June, diesel cracks have become marginally positive again for the first time since February, but gasoline cracks remain negative.

Exhibit 29: Domestic diesel and gasoline cracks turned deeply negative over March and April

China

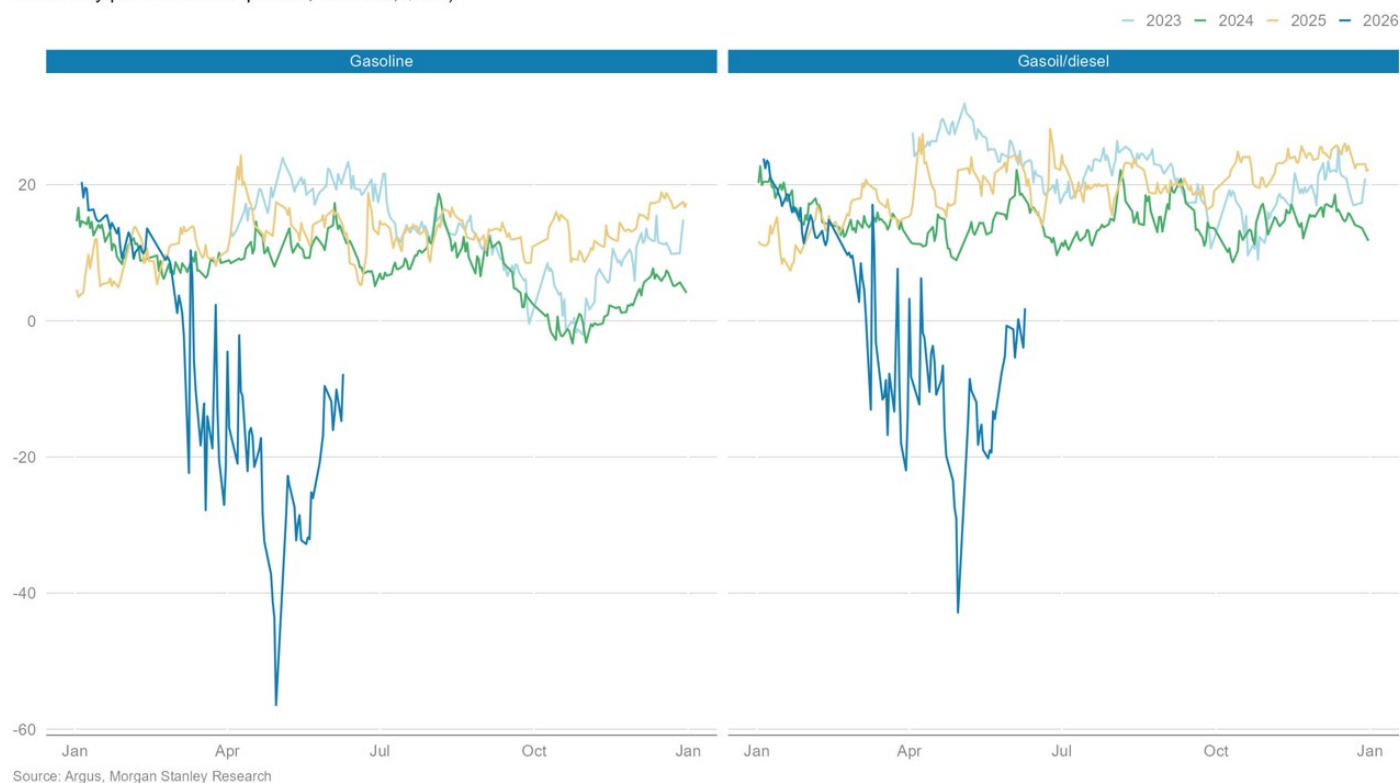
Bohai Bay product crack spreads, excl. tax, \$/bbl)



Source: Argus, Morgan Stanley Research

Exhibit 30: Seasonal changes in Chinese gasoline and diesel cracks**China**

Bohai Bay product crack spreads, excl. tax, \$/bbl)



Refinery crude throughput fell by 1.2 mb/d MoM in April to 13.35 mb/d, the lowest level since August 2022. Run levels in April were tracking ~2 mb/d lower than February levels. Chinese refiners' margins appeared strong in March, as they were still processing crude bought or hedged before the sharp rise in oil prices. However, by April more refiners were having to process higher cost crude amid a drop in Middle Eastern crude supplies. This squeezed margins for Chinese refiners in April, an issue further compounded by refined product export controls and weak domestic demand, prompting refiners to cut runs. State-owned refiners cut most aggressively, as they were the largest buyers of Middle Eastern crude and also most affected by the refined products export ban. Argus estimates that pre-conflict, ~1.5 mb/d of crude was processed to produce refined products for export, the majority of which will now have been cut. In contrast, independent refiners moderately raised runs over March and April, helped by an additional batch of time-limited crude import quotas which allowed them to bring large quantities of Russian and Iranian oil onshore, that had previously been stuck as floating storage due to OFAC sanctions restricting the pool of available buyers.

Planned refinery maintenance activity also weighed on crude throughput across March and April, as multiple state-owned refiners still undertook usual seasonal maintenance activities. Total CDU capacity taken offline was ~1.4 mb/d in April based on IIR data, still ~300 kb/d lower than this time last year when seasonal maintenance levels were very high. However, this data set does not fully reflect the lower rates of utilisation at active CDUs, meaning overall throughput was still lower YoY in April despite fewer offline CDU units.

Looking forward, Chinese refinery runs have continued to decline in May as state-owned refineries have taken more capacity offline for maintenance and continued to cut utilisation rates. Continued weak domestic refining margins are prompting refineries to take capacity offline over coming months. Sinopec took its 280 kb/d Yanzi refinery offline for maintenance in May, while mega-refiner Rongsheng has cut runs by 300 kb/d at its 800 kb/d ZPC refinery.

Exhibit 31: Chinese CDU outages rose 600 kb/d MoM in April to 1.45 mb/d

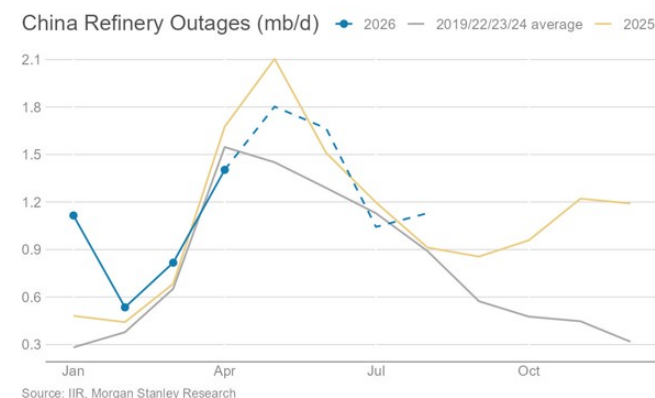
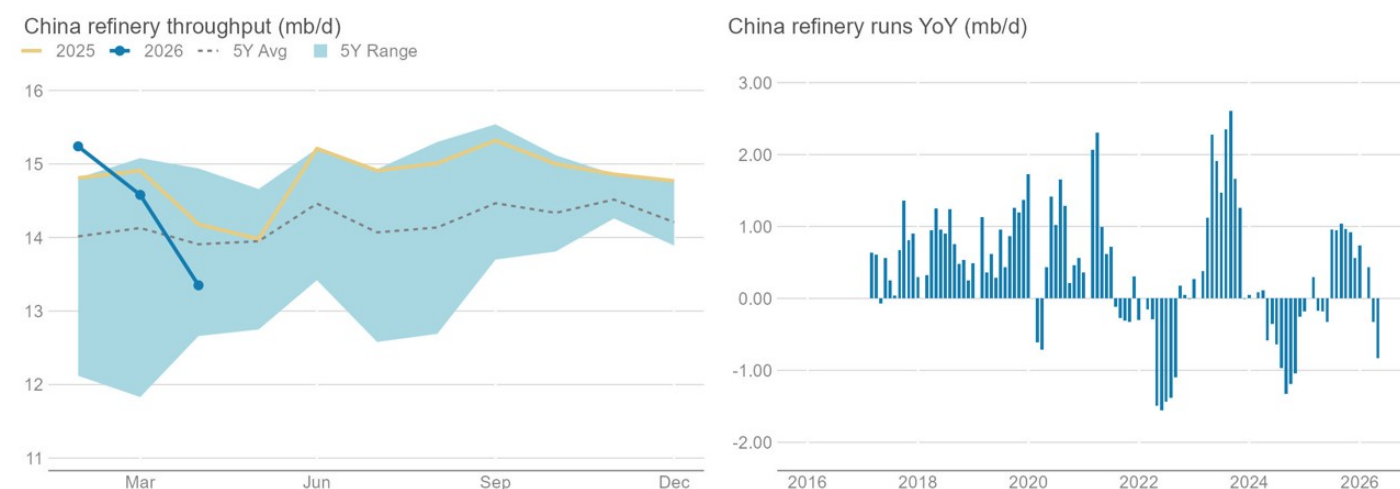


Exhibit 32: Refinery runs fell 1.2 mb/d MoM and were 830 kb/d lower YoY (-6 %) in April



State-owned

State-owned refinery run rates dropped by 3.3pp from March levels to 69%, down a full 13pp from end-February levels. Argus reports that state-owned refinery margins declined over April, averaging \$14.7/bbl in April, down from very elevated levels of \$43.3/bbl in March. State-owned refinery margins benefitted from hedging gains and higher domestic product prices over March, the effects of which faded into April. This resulted in Sinopec cutting runs by around 230 kb/d from March to 4.4 mb/d in April, while PetroChina reduced runs by about 120 kb/d to 3.9 mb/d, based on Argus data.

A decline in state-owned refinery utilisation levels is normal across March and April, as spring is usually the most intensive maintenance period of the year for the larger refiners in China. Sinopec took CDU units offline at their Tahe and Shijiazhuang refineries while CNOOC undertook planned maintenance at their Cangzhou and Zhoushan refineries.

However, the usual seasonal run rate decline for maintenance was significantly amplified by run cuts due to crude feedstock availability issues, high crude prices, refined product export ban and weak downstream demand. The government limited refiners' ability to pass through higher refined product prices to consumers, sending gasoline and diesel cracks deeply negative in April and squeezing refining margins. The export ban also prevented state-owned refiners from capitalising on attractive export margins. Finally, weak downstream sales in response to higher prices has meant that inventories of refined products actually rose at state-owned refiners

over March and April. In response to weak refining margins, state-owned firms have cut their own output and bought more transport fuels from the independent refiners, which is more attractive than refining the product themselves given negative cracks.

Looking forward, state-owned refiners continued to cut run rates in May, despite government approval of higher refined products exports, as weak refined product prices continued to depress refining margins. Stronger petrochemical margins are incentivising some yield switching away from transport fuels, and may encourage higher runs at some state-owned refiners, but overall the outlook for refining margins remains weak.

State-owned refiners have also delayed two new refinery projects due to start up in 2H 26, affecting 500 kb/d of capacity. The new 300 kb/d HAPCO refinery has also delayed its start date from 1H 26, likely until November Argus estimates, due to feedstocks supply uncertainty resulting from disrupted Strait of Hormuz flows. The project also includes a 1.65 mtpa ethylene cracker and a 2 mtpa paraxylene unit. Separately, the planned restart of a 200 kb/d crude unit at PetroChina's Dalian refinery has been postponed indefinitely, Reuters reports (8th June).

Independent

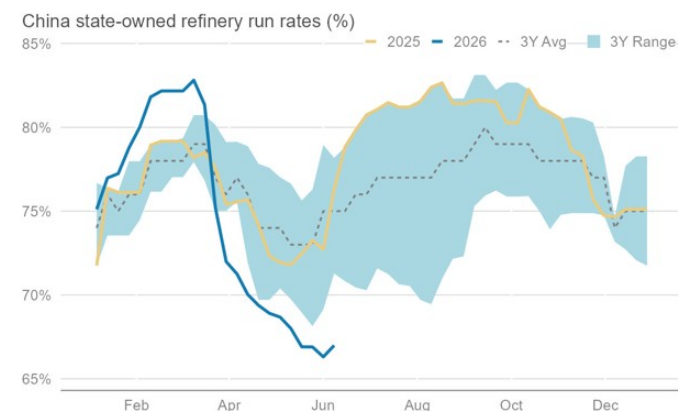
Independent refinery utilisation levels climbed by 3pp across both March and April, reaching 53% util levels at the end of March and 56% by the end of April. This marked the highest level of utilisation in China's independent refining sector since May 2024.

Independent refiners raised runs despite an unprofitable environment in March and April for Chinese independent refiners. These smaller refineries have predominantly run OFAC sanctioned crude grades over the last several years, capitalising on large discounts on Russian and Iranian oil. However, unsanctioning of both Russian and Iranian oil-on-water over March increased the pool of available buyers for these grades in Asia. Consequently, discounts on Russia's ESPO Blend, a favourite of the independent refiners, moved from a \$8.5/bbl discount to Brent pre-war to \$6-8/bbl premium in April. Similarly, Iranian Light was being offered at a \$2-3/bbl premium to Brent in April vs a \$10/bbl discount back in February. This increase in crude costs squeezed independent refining margins, turning them marginally negative over March and April.

The driver behind supportive run rates in the independent sector was instead government intervention. The government granted 50 Mt of temporary crude import quotas in early April, permitted deferred fuel tax payments and reportedly required independent refiners not to cut rates below 2025 levels (Bloomberg, April 2nd), although last year represents a low base. Argus also reported (April 14th) that officials also warned independents that failing to maintain runs could lead to cuts in future quotas. Refining margins for both state-owned and independent refiners are negative, however only independent refiners have been mandated to maintain runs, pushing the financial burden of supplying the domestic fuel market onto the independent refiners.

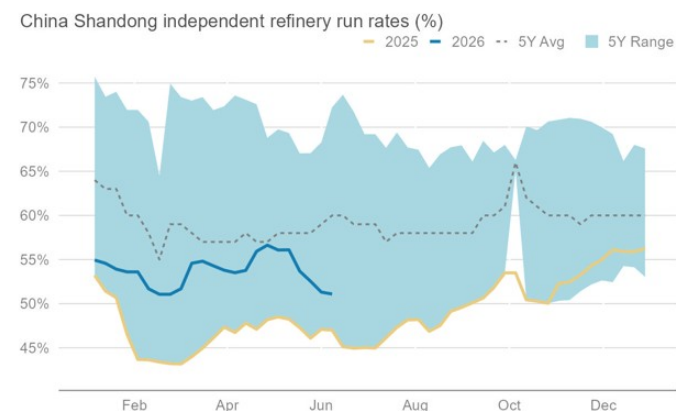
Looking forward, independent runs remained relatively stable in May despite negative margins, as the government mandated independent refiners to maintain run rates. However, independent refiners lobbied the government in May to allow them to lower runs due to negative margins and elevated inventory levels. The government then eased operating requirements, allowing refiners to cut output to no less than 80% of last year's average. This has enabled independent refiners to lower utilisation rates in June.

Exhibit 33: State-owned refineries' utilisation levels dropped sharply to 79% in March and 69% in April...



Source: Bloomberg, Morgan Stanley Research

Exhibit 34: ... with independent refinery util rates inched higher to 54% in March and 55% in April



Source: Bloomberg, Morgan Stanley Research

Run cuts from state-owned refiners resulted in a decrease in refined products output across all product types in April. Output of road fuels declined at a faster pace than output of petrochemical feedstocks, despite government orders to maximise road fuel production.

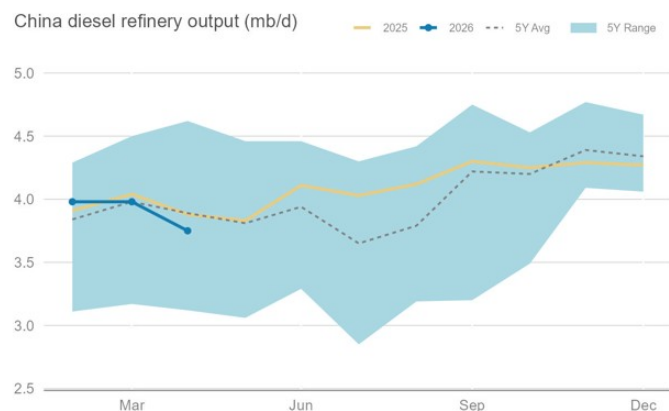
In early April, the government has ordered state-owned firms to prioritise road-fuel production over naphtha while the export controls were in force. This was in order to protect domestic transport fuel supplies. Refiners planning on raising combined gasoline and diesel yields to 56% in April, up from 49% last year. Argus reports (April 30th) that state-owned refiners Sinopec and PetroChina raised their gasoline and diesel yields by 1-5pp in April vs February.

However, the data on refinery output of refined products appears to show resilient production of naphtha and LPG in April, while output of road transport fuels fell. High refinery yields of diesel and gasoline combined with a sharp drop off in exports from mid-March created an oversupply of road transport fuels in the domestic market in early April, exacerbated by weak domestic sales. Consequently, refiners have looked to maximise petrochemical feedstocks again in the 2H of April, as chemical margins are more attractive. Sinopec increased LPG production from their refineries at the beginning of April and raised run rates at their Maoming and Zhingke steam crackers towards the end of April, while PetroChina looked to ramp up runs at their Qinzhou cracker in May. By late April state-owned refiners were then given permission from the government to increase yields of petrochemicals again, in response to spiking ethylene prices.

Refiners have also cut back on output of jet fuel, as a significant portion of China's ~150-200 kb/d of seaborne jet fuel exports have been cut under the export ban. Sinopec and PetroChina, the major jet fuel producers in China, plan to keep jet fuel output ~150 kb/d lower YoY as a result, ensuring sufficient supply for domestic market but cutting out export volumes. Overall, jet fuel output from refineries appears to be down ~200 kb/d YoY in April.

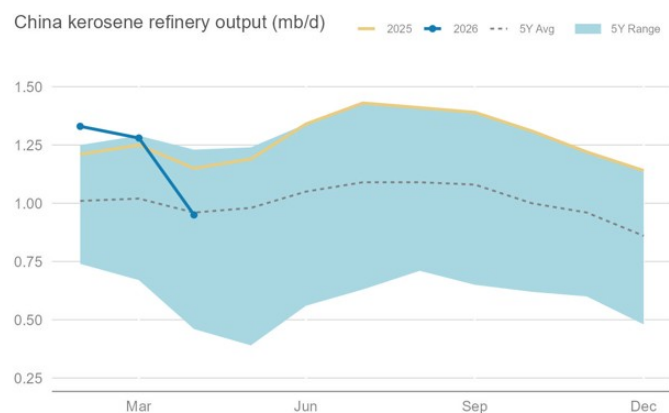
Looking forward into May, output of transport fuels will likely continue to decline while petrochemical feedstock output remains more resilient. Chinese refiners plan to cut diesel yields to ~24%, down from ~28% in May.

Exhibit 35: Refinery output of diesel fell 230 kb/d MoM and was down 130 kb/d YoY (-3%)



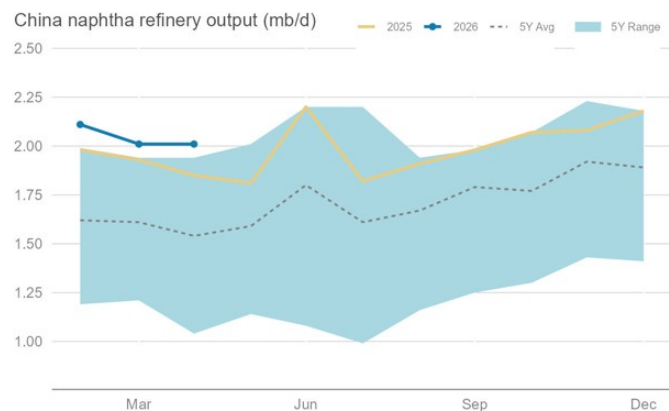
Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 37: Refinery output of jet fuel/kerosene was down 330 kb/d MoM and fell 200 kb/d YoY (-17%)



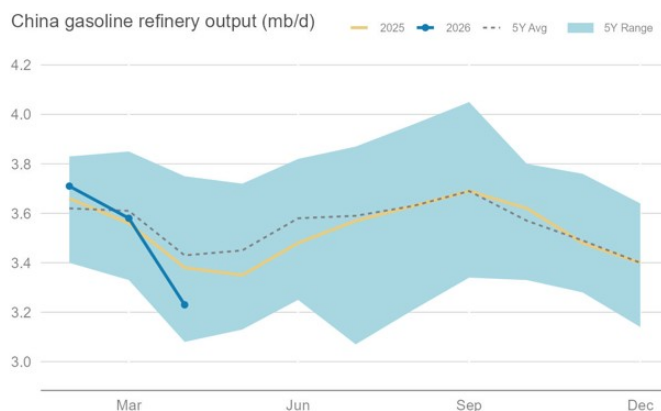
Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 39: Refinery output of naphtha was flat MoM but was 160 kb/d higher YoY (+9%)



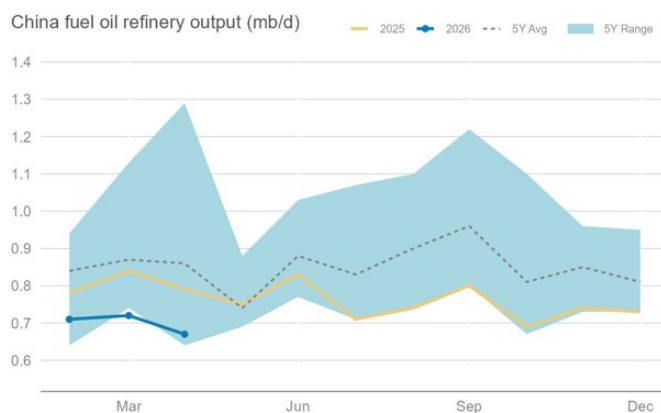
Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 36: Refinery output of gasoline was down 350 kb/d MoM and was 150 kb/d lower YoY (-4%)



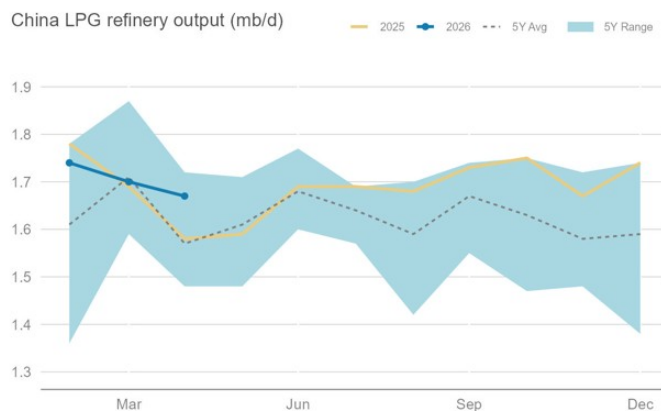
Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 38: Refinery output of fuel oil fell 5 kb/d MoM and was 120 kb/d lower YoY (-15%)



Source: China General Administration of Customs, Morgan Stanley Research

Exhibit 40: Refinery output of LPGs fell 30 kb/d MoM but was up 90 kb/d YoY (+6%)



Source: China General Administration of Customs, Morgan Stanley Research

Inventory Data

We can observe a portion of China's crude oil inventories that are held in above ground storage tanks with floating roofs, as changes in tank volume are measurable using satellite data. We call this China's observable inventories and at the end of February these totalled 1.27 billion barrels of crude. Over the course of 2025, China added ~112 mln bbls to observable inventories, implying an average build rate of 300 kb/d, with peak inventory builds of ~1.7 mb/d occurring in Dec 2025. However, note China also stores oil in underground caverns, particularly for their strategic reserves, and data visibility on these volumes is poor. The Oxford Institute for Energy Studies estimates China has around ~100 mln bbls of underground storage.

Observable crude inventories in China actually built over March and April, despite the disruption to crude supplies caused by the Middle East conflict. The Vortexa satellite data indicates China added 36 mln bbls to inventory in March and another 18 mln bbls in April, implying build rates of 1.2 mb/d and 0.6 mb/d respectively.

Despite seeming counter-intuitive, observable crude inventory builds in March and April align with the S/D balance. In March, domestic production of 4.5 mb/d plus net crude imports of 11.8 mb/d gives total crude supply of 16.3 mb/d. NBS data records refinery runs of only 14.6 mb/d, so even once allowing for some inaccuracies in the data, it would appear China's crude market was still oversupplied in March. The implied oversupply is ~1.7 mb/d, which would explain the +1.2 mb/d of observable builds plus some additional storage builds in unobservable locations. In April the S/D balance tightened as net crude imports fell to 9.3 mb/d, however significant refinery run cuts meant the implied balance was still +0.4 mb/d in surplus. This is slightly below observable crude builds of +0.6 mb/d in April, but is still consistent directionally.

However, in May, China's observable crude inventories started to draw, dropping by 23 mln bbls over the month. The government gave state-owned refiners permission to draw on commercial inventories back in April (Bloomberg, April 9th). This meant the refiners could cut back on expensive imported crude purchases and begin tapping on their inventories in May as crude inflows slowed. Vortexa data shows net crude imports dropped by another 1.7 mb/d in May. To align with the implied draw rate of 0.7 mb/d, refineries will also have had to cut runs further, likely by ~600-700 kb/d. Given continued decline in state-owned refinery utilisation rates during May and the removal of another 300 kb/d of CDU capacity for maintenance, this seems a plausible reduction in runs MoM. **Although the government has allowed commercial inventories to be tapped, the government does not appear to have authorised oil companies to access the SPR.**

Observable refined product inventories appear to have been broadly stable over March and April, however note that data coverage across refined products inventories is not very comprehensive. A combination of the refined product export ban, stable runs at independent refineries, yields shifts to diesel and gasoline and weak consumer sales appears to have supported inventory levels. Argus reports that gasoline tank utilisation at state-owned marketing companies was up 2pp MoM and 7pp YoY at end-April, with diesel tank utilisation up 4pp MoM and 6pp YoY.

Exhibit 41: China's observable crude inventories have built over March and April, as crude supply still outpaced refinery demand.

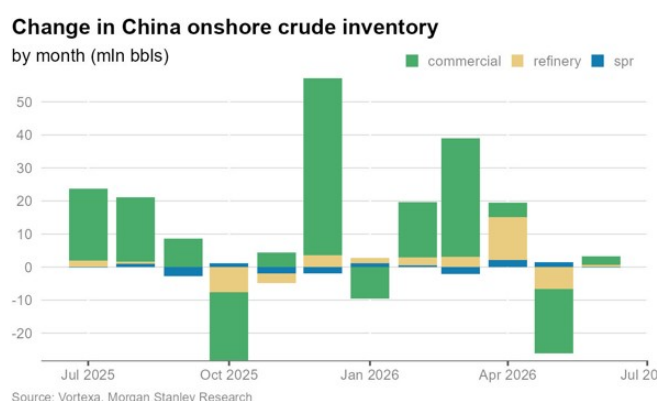


Exhibit 42: Observable crude inventories built by ~55 mln bbls over the course of March and April

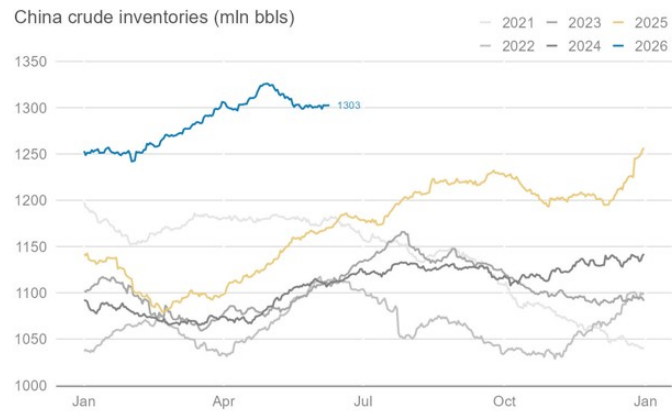


Exhibit 44: Observable gasoline inventories

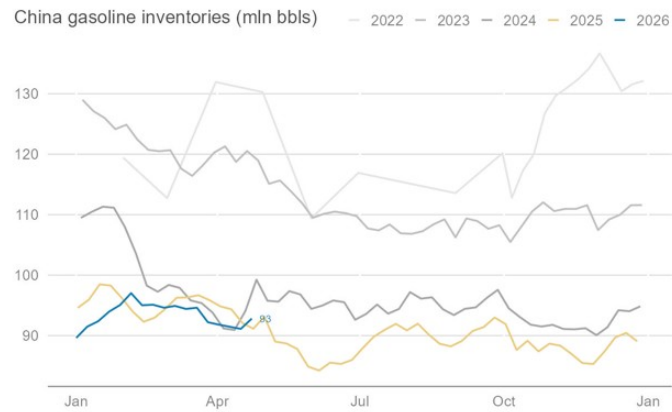


Exhibit 43: Observable aggregated product inventories

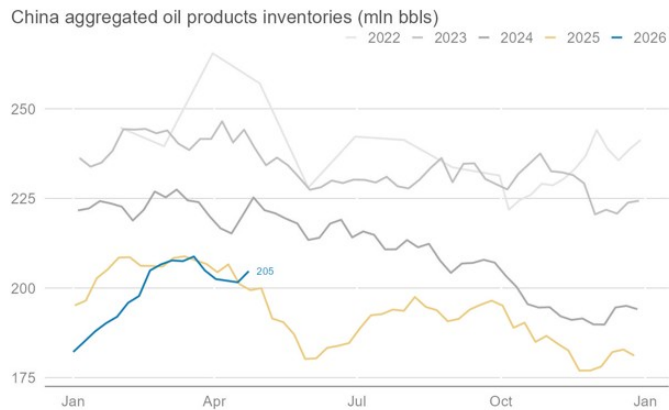
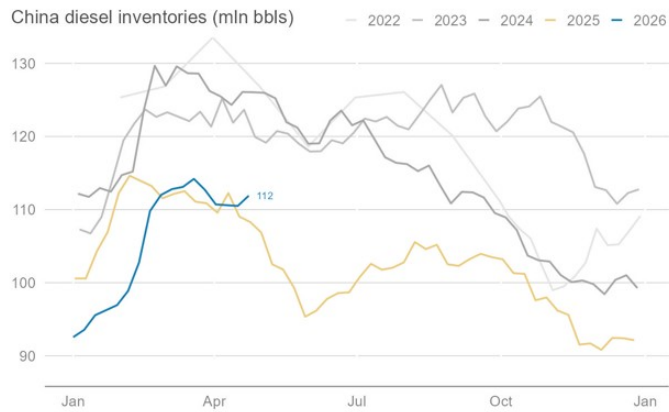


Exhibit 45: Observable diesel inventories



Comparison with other data sources

We compare seaborne crude import data from Vortexa with official reporting by China Customs for total crude imports. The customs data implies an additional ~1.1 mb/d of land/pipeline imports above the 6.7 mb/d of seaborne imports recorded by Vortexa for May. This is flat on land imports of ~1.1 mb/d in April.

In April, all data providers estimated declines in China oil demand, although with some variation on the magnitude of the decline.

S&P Global, Energy Aspects, the IEA and Argus all estimated MoM and YoY declines in China's oil demand in April 2026. At the highest end of the range, Energy Aspects sees demand down 2.0 mb/d MoM and 2.4 mb/d YoY. At the lower end of the demand contraction range, the IEA estimates only a 0.8 mb/d MoM decline and 0.3 mb/d YoY decline in China's oil demand. These estimates align broadly with our apparent demand estimates, as we observe a 1.4 mb/d decline in demand MoM and 1.2 mb/d decline in demand YoY. Taken together this equates to estimates of China oil demand contraction YoY of 2% to 14% in April.

Exhibit 46: Comparison of demand estimates from different data agencies

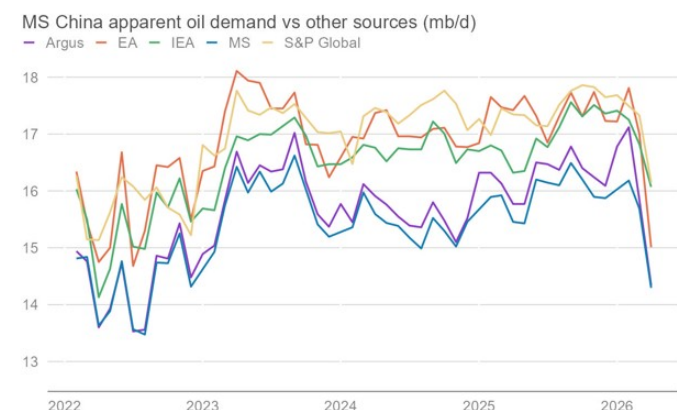
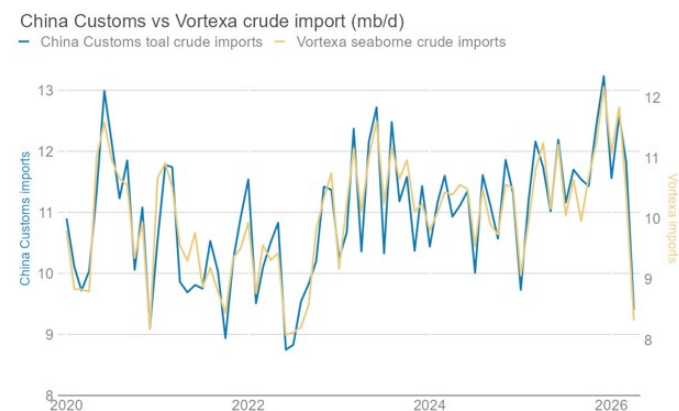


Exhibit 47: China Customs import data and Vortexa seaborne import data implies a small decline in pipeline crude imports in April



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(as of May 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1542	42%	465	51%	30%	707	43%
Equal-weight/Hold	1571	43%	369	40%	23%	723	44%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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